


**Ved Patel**
Course : **Bachelor of Engineering, Mechanical Engineering, 2028**
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CGPA : 8.49/10


**ACADEMIC DETAILS**

| COURSE | SPECIALISATION | INSTITUTE | BOARD | SCORE | YEAR |
|---|---|---|---|---|---|
| CLASS XII | SCIENCE/MATHS | Green Valley High School, Vadodara | CBSE | 94.6% | 2023-24 |
| CLASS X | GENERAL | Navrachana School Sama, Vadodara | CBSE | 92.7% | 2021-22 |

## CORE AND ELECTIVE COURSES

**Core Courses:** Advanced Mechanics of Solids, Fluid Mechanics, Applied Thermodynamics, Material Science and Engineering, Heat Transfer, Design of Machine Elements, Advanced Manufacturing Processes, Mechanisms and Machines, Manufacturing Management, Engines Motors and Mobility

**Electives**(including Robotics Minor)**:** Robotics, Artificial Intelligence for Robotics, Finite Element Method, Internet of Things

**Humanity Electives:** Science Technology and Modernity, Cinematic Adaptation

## INTERNSHIPS AND PROFESSIONAL EXPERIENCE

### Summer Intern - DRDO CEMILAC, Bengaluru — May 2026 - July 2026

*Centre for Military Airworthiness and Certification, a premier aerospace research and certification laboratory under DRDO*

- Developed a **2D incompressible fluid flow solver** from scratch in MATLAB using the **Finite Difference Method** for aerodynamic modeling.
- Bypassed commercial CFD software to save **₹50L+** in licensing costs while retaining absolute control over all simulation parameters
- Successfully captured, simulated, and analyzed complex transient aerodynamic flow phenomena, such as **von Kármán vortex shedding**
- Completed a comprehensive, **expert-led training** course on **CFD++** simulation software directly from the company's representatives

### Design Core, Radio Control Club — January 2025 - Present

*BITS Pilani's premier aeromodelling club dedicated to the engineering, fabrication, and flight-testing of custom unmanned aircraft*

- Engineered scale RC aircraft **(B-2, F-22)** and optimized custom airfoils using **XFLR5** to simulate diverse structural geometries
- Fabricated lightweight airframes using **carbon fiber** and epoxy layups, alongside composite builds of Depron, foam, and balsa wood
- Integrated flight controllers and autonav systems, mastering complex multi-axis RC flight dynamics via rigorous **PicaSim** simulations

## POSITION OF RESPONSIBILITY

### System Engineer, Mechanical Subsystem, CRISS Robotics — February 2026 - Present

*BITS Pilani's student-led engineering team developing autonomous Mars rovers for extreme environments and international competitions*

- Engineered and assembled rugged mechanical frameworks and chassis for an autonomous Mars rover prototype using **SolidWorks** and **Fusion 360**
- Validated structural durability, fatigue life, and load capacity through rigorous **ANSYS simulations** for high-stress and extreme environments
- Helped secure 2nd in Asia at **ERC** 2025, a Top 10 finish at the **International Rover Challenge**, and being the only Indian team invited to **ARCh** 2026

### Coordinator, Department of Publications and Correspondence, APOGEE — April 2026 - Present

*Department of Publications and Correspondence for BITS Pilani's technical festival, APOGEE*

- Spearheaded external **correspondence** and strategic inter-college outreach campaigns, driving a record external footfall of **1,000+ students**
- Conducted a college-wide club photography and printing drive, raising **₹25L+ in revenue** to significantly fund the technical fest budget
- Managed campus-wide **publication** requests, press releases, and print promotional material distribution across all organizing departments

### Treasurer, Gurjari Association — April 2026 - Present

*The Gujarati cultural association for the campus, hosting Garba Nite, a student-run event and the largest cultural event by an association on campus*

- Directed a **₹2L event budget** and **independently** managed all financial operations, vendor transactions, and cash flows for the annual Garba Nite
- **Generated ₹3L+** in gross revenue by executing strategic ticketing sales campaigns and securing high-value corporate **sponsorships**
- Successfully orchestrated event logistics, vendor coordination, and hospitality operations, hosting a record footfall of **3,000+ attendees**

## VOLUNTEERING AND ACHIEVEMENTS

### Umang Head, National Service Scheme (NSS)

Orchestrated donation drives across students, faculty, and alumni, raising **₹16L+** over two years to sponsor **150+ underprivileged students.**

### School Prefect, Navrachana School

Served as student **council leader**, maintaining discipline and coordinating large-scale events while liaising between students and administration.

## SKILLS

**SKILLS:** Fusion 360, Solidworks, ANSYS, CFD++, XFLR5, MATLAB, Simulink, C Programming, Python, Arduino IDE, Bamboo Studio, Orca, AutoCAD